

Jiawei Lian, Curriculum Vitae

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-**Birth Date:** 01/1997

-**Status:** Final-year PhD student, PolyU and NWPU joint training leading to dual PhD

-**Research Interest:** AI Safety, Large language model, Computer vision, Remote sensing

-**Postal Address:** Room 505, Building CF, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong

Education

- 01/2024 – Present  Ph.D., Artificial Intelligence and Signal Processing, Advisor: [Lap-Pui Chau](#)
The Hong Kong Polytechnic University (PolyU)
Thesis: *Towards Exploring the Adversarial Robustness of Deep Learning Models*
- 04/2022 – Present  Ph.D., Information and Communication Engineering, Advisor: [Shaohui Mei](#)
Northwestern Polytechnical University (NWPU)
Thesis: *Research on Adversarial Attacks against Object Detection Models*
- 09/2019 – 03/2022  M.E., Control Engineering, Advisor: [Junhong He](#)
Northwestern Polytechnical University
Thesis: *Research on Rapid Detection of Surface Defects Based on Deep Learning*
- 09/2015 – 07/2019  B.E., Automation, Advisor: Junlin Zhu
JiangXi University of Science and Technology
Thesis: *Design of 4R Joint Robot Control System based on PROFIBUS and T-CPU*

Research Publications

Journal Articles

- [1] **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Shaohui Mei, Lap-Pui Chau. “PADetBench: Towards Benchmarking Physical Attacks against Object Detection.” *Knowledge-Based Systems*, 2025.
- [2] Shaohui Mei, **Jiawei Lian**, Xiaofei Wang, Yuru Su, Mingyang Ma, Lap-Pui Chau. “A Comprehensive Study on the Robustness of Image Classification and Object Detection in Remote Sensing: Surveying and Benchmarking.” *Journal of Remote Sensing*, 2024. (The first author is the Ph.D. advisor)
- [3] **Jiawei Lian**, Xiaofei Wang, Yuru Su, Mingyang Ma, Shaohui Mei. “CBA: Contextual Background Attack against Optical Aerial Detection in the Physical World.” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–16, 2023.
- [4] **Jiawei Lian**, Shaohui Mei, Shun Zhang, Mingyang Ma. “Benchmarking Adversarial Patch against Aerial Detection.” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–16, 2022.
- [5] **Jiawei Lian**, Junhong He, Yun Niu, Tianze Wang. “Fast and Accurate Detection of Surface Defect Based on Improved YOLOv4.” *Assembly Automation*, vol. 42, no. 1, pp. 134–146, 2021.
- [6] **Jiawei Lian**, Junhong He, Yun Niu, Tianze Wang. “Rapid Detection of Surface Defects Based on Multi-scale Compression CNN.” *Computer Integrated Manufacturing System*, vol. 28, no. 11, pp. 3624, 2021.

Conference Papers

- [1] **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Shaohui Mei, Lap-Pui Chau. “Semantic Representation Attack against Aligned Large Language Models.” *Advances in Neural Information Processing Systems (NeurIPS)*,

2025.

[2] **Jiawei Lian**, Xiaofei Wang, Yuru Su, Mingyang Ma, Shaohui Mei. "Contextual Adversarial Attack against Aerial Detection in the Physical World." *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pp. 6632–6635, 2023.

Preprints

[1] **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Xiaofei Wang, Yingjie Lu, Shaohui Mei, Lap-Pui Chau. "Revealing the Intrinsic Ethical Vulnerability of Aligned Large Language Models." *arXiv preprint arXiv:2504.05050*, 2025. **(Under second round review, Nature Communications)**

[2] **Jiawei Lian**, Shaohui Mei, Xiaofei Wang, Yi Wang, Lefan Wang, Yingjie Lu, Mingyang Ma, Lap-Pui Chau. "Attack Anything: Blind DNNs via Universal Background Adversarial Attack." *arXiv preprint arXiv:2409.00029*, 2024. **(Under review, IEEE Transactions on Dependable and Secure Computing)**

China Patents

- 16/05/2023  A method and device for generating adversarial camouflage patterns against object detection. (Patent acceptance number: 202310547602.8)
- 16/12/2022  A method and device for physical attack-oriented self-adaptive adversarial patch generation. (Patent acceptance number: 202211617496.8)
- 15/12/2022  A method and device for generating adversarial patches for unobstructed physical adversarial attacks. (Patent acceptance number: 202211612346.8)
- 05/06/2021  A fast surface defect detection method based on the lightweight convolutional neural network. (Patent acceptance number: 202110627893.2)

Miscellaneous Experience

Certifications

- 04/2022  **IELTS: Overall band score 7.0 (Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0)**
- 07/2017  **Fitness Coach Licentiate (FCL): Awarded by The Ministry of Human Resource and Social Security, P.R. China.**

Involved Projects

- 03/2024-Present  Research on intrinsic ethical vulnerability and semantic representation attack against aligned large language models.
- 09/2022-02/2024  Research on the robustness of visual perception based on deep learning.
- 12/2021-08/2022  Research on background adversarial perturbations for physical attack. **Relevant research achievements received an interview invitation from the British authoritative magazine New Scientist.**

Internship

- 07/2021-09/2021  Internship at AVIC Xi'an Flight Automatic Control Research Institute.
- 07/2020-09/2020  Internship at Xi'an Institute of Modern Control Technology.